Topic 12 — Gravitational Fields

12.1 Newton's Law of Gravitation

Newton's law of universal gravitation: Every particle in the universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres.

$$F = GMm / r^2$$

where:

- **F** = gravitational force (N) — always attractive
- **G** = universal gravitational constant = 6.674×10^{-11} N m^2 kg^{-2}
- **M, m** = masses of the two objects (kg)
- **r** = distance between the centres of the masses (m)

The negative sign is sometimes included ($F = -GMm/r^2$) to indicate the force is **attractive** (directed towards the other mass). Gravity is always attractive — there is no gravitational repulsion.

Exam Tip: The distance r is measured between the **centres** of the objects, not their surfaces. If given the radius of the Earth and the altitude of a satellite, you must add them: $r = R_E + h$. This is one of the most common errors in gravitational field calculations.

Worked Example: Calculate the gravitational force between the Earth ($M = 5.97 \times 10^{24}$ kg) and a 70 kg person standing on its surface ($R_E = 6.37 \times 10^6$ m).

$F = GMm/r^2 = (6.674 \times 10^{-11} \times 5.97 \times 10^{24} \times 70) / (6.37 \times 10^6)^2$

$F = (2.789 \times 10^{16}) / (4.058 \times 10^{13})$ = **687 N**

This equals the person's weight: $W = mg = 70 \times 9.81 = 687$ N ✓

12.2 Gravitational Field Strength

Gravitational field strength (g): The gravitational force per unit mass experienced by a small test mass placed at that point in the field. It is a vector quantity pointing towards the mass creating the field.

$$g = F / m$$

For a point mass or uniform sphere (outside the surface):

$$g = GM / r^2$$

Units: N kg^{-1} (equivalent to m s^{-2}). At the Earth's surface: $g \approx 9.81$ N kg^{-1}.

Key features of g:

- g decreases with distance from the mass according to an inverse-square law
- g at the surface of a planet depends on both the planet's mass and radius: $g = GM/R^2$
- Inside a uniform sphere, g decreases linearly with distance from the centre (g = 0 at the centre)
- g is independent of the test mass — all objects experience the same gravitational field strength

Worked Example: Calculate the gravitational field strength on the surface of Mars ($M = 6.42 \times 10^{23}$ kg, $R = 3.39 \times 10^{6}$ m).

$g = GM/R^2 = (6.674 \times 10^{-11} \times 6.42 \times 10^{23}) / (3.39 \times 10^{6})^2$
$g = (4.285 \times 10^{13}) / (1.150 \times 10^{13})$ = **3.73 N kg^{-1}**

This is about 38% of Earth's surface gravity. A 70 kg astronaut would weigh $70 \times 3.73 = 261$ N on Mars, compared to 687 N on Earth.

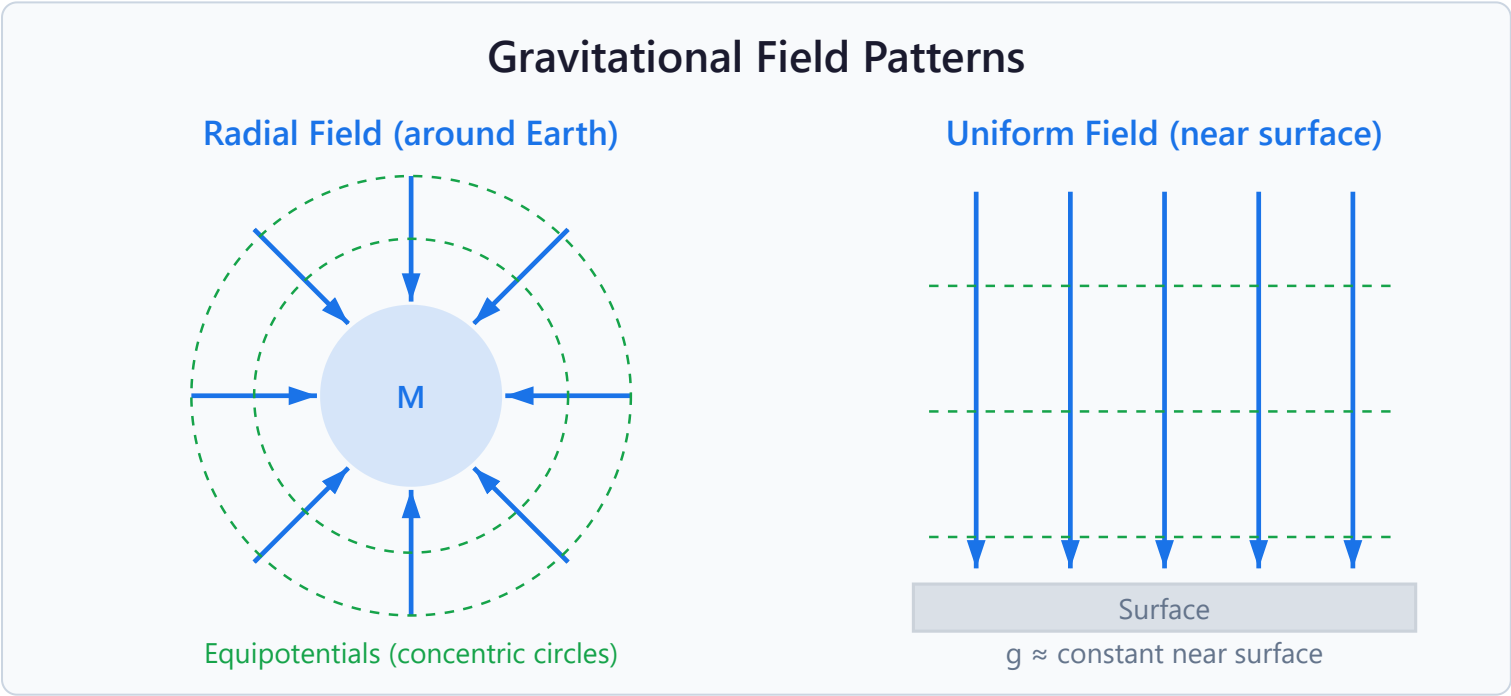
12.3 Gravitational Field Patterns

Radial Fields

- Field lines point **radially inward** towards the centre of the mass
- Field strength decreases with distance (inverse square law)
- Produced by a point mass or a spherical mass (outside the sphere)
- Field lines are not parallel — they converge at the centre

Uniform Fields

- Field lines are **parallel and equally spaced**
- Field strength is constant at all points
- A good approximation for the gravitational field near the Earth's surface over small regions (where the curvature is negligible)



Equipotentials

- **Equipotential lines** (or surfaces) join points of equal gravitational potential
- Equipotentials are always **perpendicular** to field lines
- No work is done moving a mass along an equipotential surface

- For a radial field, equipotentials are concentric circles (spheres in 3D) centred on the mass — they are closer together near the mass where the field is stronger
- For a uniform field, equipotentials are equally spaced horizontal lines

12.4 Gravitational Potential

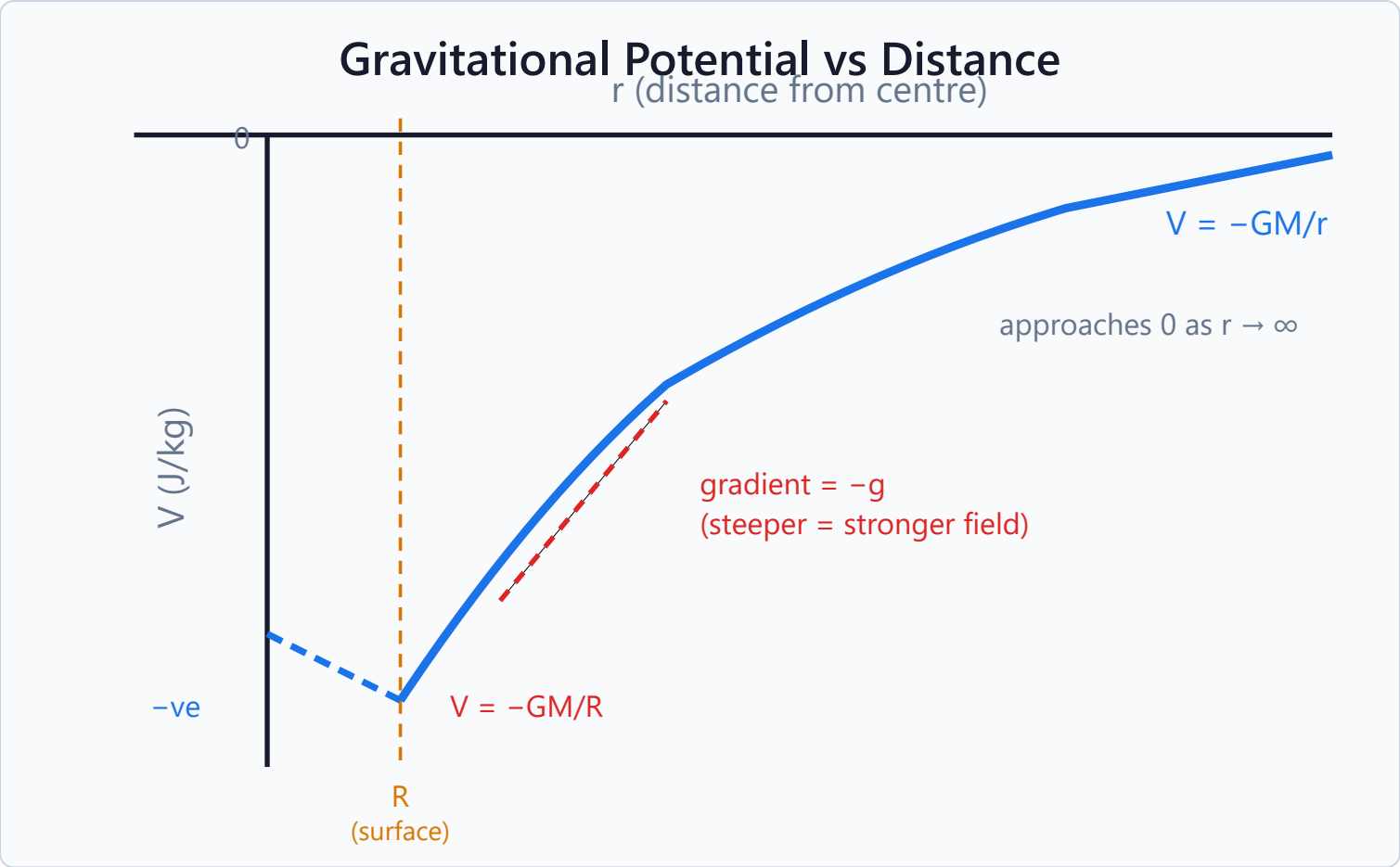
Gravitational potential (V): The work done per unit mass in bringing a small test mass from infinity to that point in the field. It is a scalar quantity and is always negative (since gravity is attractive and we define V = 0 at infinity).

$$V = -GM / r$$

Units: J kg^{-1}

Key points:

- V is always **negative** (or zero at infinity)
- V becomes more negative as you approach the mass (you go deeper into the "gravitational well")
- The potential at infinity is defined as zero
- V follows a 1/r relationship (not $1/r^2$ like field strength)



Relationship Between g and V

$$g = -dV/dr$$

The gravitational field strength equals the negative gradient of the potential. On a V–r graph, g is the magnitude of the gradient at any point. The negative sign means g points in the direction of decreasing potential (towards the mass).

Exam Tip: On a graph of V against r, the gradient gives −g. A steeper gradient means a stronger field. At the surface of a planet, the gradient is steepest. Compare with the electric field: E = −dV/dr. The mathematics is identical.

12.5 Gravitational Potential Energy

Gravitational potential energy (E_p): The work done in bringing a mass m from infinity to that point in the gravitational field of mass M.

$$E_p = -GMm / r$$

Note: $E_p = mV$. The GPE is negative because work must be done *against* gravity to move an object from that point to infinity (i.e., the object is bound to the gravitational field).

The **change** in GPE when moving between two points:

$$\Delta E_p = GMm(1/r_1 - 1/r_2)$$

where $r_2 > r_1$ (moving away from the mass).

Exam Tip: Do not confuse $E_p = mgh$ (valid only for uniform fields near the surface, where $h << R$) with $E_p = -GMm/r$ (the general expression valid at any distance). Use the general form for orbital and escape velocity problems.

Worked Example: Calculate the energy needed to move a 500 kg satellite from an orbit of radius 7000 km to an orbit of radius 10 000 km around the Earth ($M = 5.97 \times 10^{24}$ kg).

$\Delta E_p = GMm(1/r_1 - 1/r_2)$
$= 6.674 \times 10^{-11} \times 5.97 \times 10^{24} \times 500 \times (1/7.0\times10^6 - 1/1.0\times10^7)$
$= 1.991 \times 10^{16} \times (1.429\times10^{-7} - 1.000\times10^{-7})$
$= 1.991 \times 10^{16} \times 4.29 \times 10^{-8}$
$=$ **8.54×10^8 J ≈ 854 MJ**

Note: This is just the change in PE. The total energy change for the orbital transfer also involves a change in kinetic energy.

12.6 Escape Velocity

Escape velocity: The minimum speed an object must have at the surface to escape to infinity without any further energy input (i.e., to reach infinity with zero kinetic energy and zero potential energy).

Setting total energy at surface = total energy at infinity (= 0):

$$\tfrac{1}{2}mv^2 + (-GMm/r) = 0$$

$$v_{escape} = \sqrt{(2GM / r)}$$

Key points:

- Escape velocity does **not** depend on the mass of the escaping object — only on M and r of the body being escaped from
- Escape velocity = √2 × orbital velocity at that radius
- For Earth: $v_{escape} \approx 11.2$ km s^{-1}
- If an object has speed less than v_{escape}, it will return (or go into orbit if launched tangentially)

Worked Example: Calculate the escape velocity from the Moon's surface ($M = 7.35 \times 10^{22}$ kg, $R = 1.74 \times 10^6$ m).

$v_{escape} = \sqrt{(2GM/R)} = \sqrt{(2 \times 6.674\times10^{-11} \times 7.35\times10^{22} / 1.74\times10^6)}$
$= \sqrt{(5.63\times10^6)} =$ **2370 m s^{-1} ≈ 2.4 km s^{-1}**

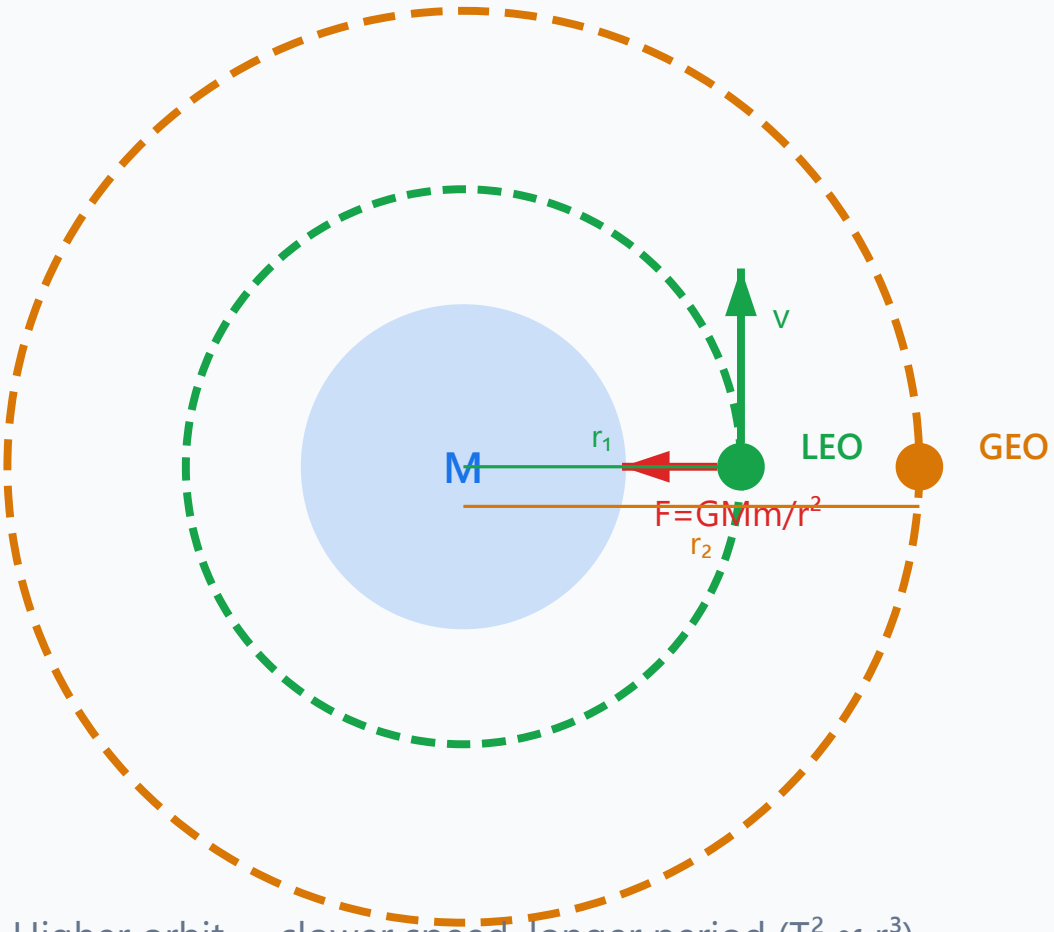
This is much less than Earth's 11.2 km s^{-1}, which is why the Moon has essentially no atmosphere — gas molecules easily reach escape velocity due to thermal motion.

12.7 Orbital Motion

For a satellite in a circular orbit, gravitational force provides the centripetal force:

$$GMm / r^2 = mv^2 / r$$

Satellite Orbital Mechanics



Higher orbit → slower speed, longer period ($T^2 \propto r^3$)

Orbital Speed

$$v = \sqrt{(GM / r)}$$

Orbital speed depends only on the mass of the central body and the orbital radius — not on the satellite's mass. A higher orbit means a **slower** orbital speed.

Orbital Period

Since $v = 2\pi r / T$:

$$T = 2\pi r / v = 2\pi \sqrt{(r^3 / GM)}$$

Squaring:

$$T^2 = (4\pi^2 / GM)\, r^3$$

This is **Kepler's third law**: T^2 is proportional to r^3. A graph of T^2 vs r^3 is a straight line through the origin with gradient $4\pi^2/(GM)$.

Worked Example: Calculate the orbital radius and speed of a geostationary satellite (T = 24 hours = 86 400 s, $M_{Earth} = 5.97 \times 10^{24}$ kg).

From $T^2 = (4\pi^2/GM)r^3$:
$r^3 = GMT^2/(4\pi^2) = (6.674\times10^{-11} \times 5.97\times10^{24} \times (86400)^2) / (4\pi^2)$
$r^3 = (2.975\times10^{24}) / 39.48 = 7.534 \times 10^{22}$
$r = (7.534 \times 10^{22})^{1/3} =$ **4.22×10^7 m ≈ 42 200 km** from Earth's centre

Altitude above surface = 42 200 − 6370 = **35 800 km**

Speed: $v = 2\pi r/T = 2\pi \times 4.22\times10^7 / 86400 =$ **$3070\ m\ s^{-1} \approx 3.1\ km\ s^{-1}$**

12.8 Kepler's Laws

Law	Statement
First Law	All planets orbit the Sun in **ellipses** with the Sun at one focus.
Second Law	A line joining a planet to the Sun sweeps out **equal areas in equal time intervals**. (The planet moves faster when closer to the Sun.)
Third Law	The square of the orbital period is proportional to the cube of the semi-major axis (mean orbital radius): **$T^2 \propto r^3$**.

Exam Tip: Kepler's third law can be verified experimentally by plotting T^2 against r^3 — the result is a straight line through the origin with gradient $4\pi^2/(GM)$. This allows the mass M of the central body to be determined. This technique is used to measure the mass of planets (from their moons' orbits), stars (from orbiting planets), and even black holes.

12.9 Types of Orbit

Geostationary Orbit

- Orbital period = **24 hours** (same as Earth's rotation)
- Orbits directly above the **equator** (inclination = 0°)
- Same direction as Earth's rotation (west to east)
- Appears **stationary** relative to a point on the Earth's surface
- Altitude ≈ 35 800 km above the surface (r ≈ 42 200 km from centre)
- **Uses:** Communications satellites, TV broadcasting, weather monitoring

Low Earth Orbit (LEO)

- Altitude: typically 200–2000 km above the surface
- Orbital period: ~90 minutes to 2 hours
- Can be at any inclination (including polar orbits at ~90°)
- Satellites move relative to the Earth's surface
- **Uses:** Earth observation, scientific research, International Space Station, imaging/reconnaissance, Starlink internet constellation

Property	Geostationary	Low Earth Orbit
Altitude	~35 800 km	200–2000 km
Period	24 hours	~90 min – 2 hours
Orbital speed	~3.1 km/s	~7.5–8 km/s
Inclination	0° (equatorial)	Any (including polar)
Position relative to surface	Stationary	Moves across the sky
Coverage	Sees 1/3 of Earth (3 satellites cover globe)	Sees small area (needs many satellites)

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Exam Tip: A higher orbit means a lower orbital speed but a longer orbital period. Do not confuse the two — students often incorrectly think faster orbit means shorter period at all radii. Use $v = \sqrt{(GM/r)}$ to confirm that v decreases as r increases, and $T = 2\pi\sqrt{(r^3/GM)}$ to confirm that T increases as r increases.

Total Energy of a Satellite

For a satellite in a circular orbit:

- Kinetic energy: $E_k = ½mv^2 = GMm/(2r)$
- Potential energy: $E_p = -GMm/r$
- Total energy: $E_{total} = E_k + E_p = -GMm/(2r)$

The total energy is **negative** (the satellite is bound). Moving to a higher orbit increases E_{total} (makes it less negative), which requires energy input. The kinetic energy decreases (slower speed) but the potential energy increases by twice as much.